

3G Network and Security

Module 42

Duration: 25:49

Video: <https://www.youtube.com/watch?v=wdT6MOpPhcs>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala

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0:05 [Music]

0:12 [Applause]

0:21 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:30 we will study about 3G network and 3G network security history of 3G network

0:37 3G technology is the result of research and development work carried out by International telecommunication Union

0:44 it in the early 1980s 3G specifications and standards were developed in 15 years the technical

0:53 specifications were made available to the public under the name IMT 2000 the communication spectrum between 400 MHz

1:02 to 3 GHz was allocated for 3G the first pre-commercial 3G network

1:09 was launched by NTT DoCoMo in Japan in 1998 branded as FOMA F4 the first

1:19 European pre-commercial network was an UMTS Network on the island of Man by

1:26 Telecom the first Network to go commercially live was by SK Telecom in South Korea on the CDMA based one

1:37 EV-DO technology in January 2002 the first commercial United States

1:44 3G network was by Verizon mobile networks on CDMA 2001 into EV-DO technology the first

1:54 pre-commercial demonstration Network in the southern hemisphere was built in Adelaide South Australia by

2:02 Optusnet Corporation in February 2002 using UMTS on 2100

2:10 MHz Optusnet launched the first 3G Network in Africa 11th December 2008 India

2:19 entered the 3G arena with the launch of 3G enabled mobile and data services by

2:25 government owned Mahanagar Telephone Nigam Limited (MTNL) in Delhi and later in

2:32 Mumbai MTNL becomes the first 3G mobile service provider in India after MTNL

2:39 another state operator BSNL launched 3G services on

22nd Feb 2009 in Chennai and later launched 3G as Nationwide however before the rise of the 3G Evolution a transition was made from GPRS to Edge or 3 minutes, 1 second egprs enhanced GPRS which means GPRS on a Red Bull like high MMS boomed and mobile internet took off into the stratosphere this brings us to 3G technically it's a direct upgrade to Edge networks but it runs on a wholly different set of frequencies entirely new infrastructure has to be deployed for the successful roll out of 3G what is 3G network 3G is the third generation of Wireless Mobile telecommunications technology this is based on a set of Standards used for mobile devices and mobile telecommunications user services and networks that comply with the international mobile telecommunications 2,000 that is IMT 2000 specifications by the international telecommunication Union 3G means faster connectivity to the internet via your mobile device the chance to finally have a video call while traveling by train or standing on the beach and enjoying the sunshine sending images videos emails Etc as fast as it can be done via desktop computer the third generation 3G mobile devices and services will transform Wireless Communications into online realtime connectivity 3G Wireless technology allows an individual to have immediate access to location specified services that offer information on demand mobile phones are rapidly becoming the preferred means of personal communication creating the world's largest consumer electronics Industry 3G finds application in Wireless voice telephony mobile internet access fixed wireless internet access video calls and Mobile TV 3G is the result of development and evolution of GSM standards 3G telecommunication networks support services that provide an information transfer rate of at least 200 kilobit per second later 3G releases often denoted 3.5g and 3.75g also provide mobile broadband excess of several mbits per second to smartphones and mobile mods in laptop computer s this ensures it can be applied to wireless voice telephony mobile internet access fixed wireless internet access video calls and Mobile TV Technologies figure shows the various Technologies related to 3G and

their suitability for mobile users

- 5:48 Beyond 3D it is predicted that mobile networks and the wireless communication landscape will be based on various
- 5:55 Technologies offering seamless Mobility with cellular network networks fourth generation 4G services will enable
- 6:04 Broadband wireless communication at home at the office and on the Move in other words new networks will make the
- 6:11 services provided by the web and the internet as well as a variety of other services such as multimedia and
- 6:19 entertainment available to mobile users 3D capabilities highspeed data
- 6:26 Transmissions symmetrical and asymmetrical data transmission support improved voice
- 6:33 quality greater capacity multiple simultaneous Services global roaming across networks
- 6:43 improved Security Service flexibility 3G applications wireless internet audio on
- 6:51 demand electronic postcards video conferencing secure mobile Commerce transactions
- 7:00 traffic and traveling information location specified 3G Services Services
- 7:07 provided by 3D Network in general terms 3D Services comprise of wide bandwidth service such as enhanced communication
- 7:15 example messaging email video browsing the web and location specific information like informing users about
- 7:24 the availability of stores restaurants gas stations free parking Lots so on
- 7:30 near them in addition business users will have a direct access to company networks while traveling or working
- 7:38 outside office from marketing point of view identifying and designing these services and setting an appropriate
- 7:45 price is said to be the core marketing task related to 3G Mobile internet browsing the web from
- 7:55 mobile the aim to merge cellular networks and the internet meaning that mobile users can have access to all
- 8:03 services that the internet provides from messaging to contemporary management research the term mobile internet or
- 8:11 internet in Mobile refers to gaining access to the internet using an handheld wireless device like a mobile phone as
- 8:19 shown in the figure or PD as 2G networks have been mainly voice Centric with low
- 8:25 data transmission capacity 2.5g and 3G will speed up data transmission speeds
- 8:33 however in 3G networks the data transmission speed is dependent upon the number of users accessing the network at

8:41 the same point of time email Services email service was rated as the number one preferred mobile service by Sweden

8:50 in 2001 followed by Banking and use of Encyclopedia email Services can be

8:57 categorized into web style HDM ml email service and plain text email Services

9:03 HTML email Services provide more flexibility with the format and appearance while plain text email

9:10 comprises of a letter style message although email has become a widely acceptable and accessible Communications

9:18 medium during the recent years there is lack of research about the use of email from mobile devices for many internet

9:26 users both private and business users email has been the most common way of exchanging information documents and

9:35 communication although email is cheap easy to access and is asynchronous in nature email Services have suffered from

9:44 the receipt of unsolicited messages or even spam the fear of receiving unsolicited messages

9:52 and spam on mobile devices may slow the adoption of email Services among mobile users second

9:59 as the first mobile virus news has been reported in media BBC News 2003 mobile users might want to protect their mobile

10:08 devices by avoiding using email Services messaging services mobile messaging

10:16 referring to short message service SMS and multimedia message service MMS is expected to be the most utilized mobile

10:25 service in the future it is estimated that the step by by step consumers as well as business users will upgrade

10:32 their messaging from sending simple text messages to multimedia messages including pictures and video clips

10:40 although 3G networks will make bandwidth hungry services such as sending receiving and downloading video clips

10:48 possible there are notable cultural differences in adopting multimedia messaging services for example while the

10:57 European Market has not adopted multimedia messaging to large extent due to various reasons such as pricing

11:04 policy and relatively complex usage Asians have eagerly adopted MMS on average while a typical user in Europe

11:13 sends one to two MMS messages per month and Asian takes and sends 20 to 30 MMS

11:20 messages besides SMS and MMS Services instant messaging I am on the Internet is one of the most popular services

11:29 especially among youngsters the instant messaging service has the potential to be
utilized widely among mobile users as the service allow
11:38 users to send content example a text message HTML page a picture a file
containing a song a video clip or other
11:47 file to another user in near real time this instant messaging service is combined
with presence service giving
11:56 the possibility to see if other users are available and reachable the framework in
figure shows some of the
12:04 major Services provided by 3G and underlying factors 3G architecture 3G wireless
networks
12:13 consist of a radio Access Network R and a core Network the core Network consists
12:20 of a packet switch domain which includes 3G ggsn and
12:27 ggsn which provide the same functionality that they provide in a GPRS system and
a circuit switch domain
12:36 which includes 3G MSC for switching off voice calls charging for services and
12:43 exess is done through the charging Gateway function cgf which is also part
12:50 of the core Network AR and functionality is independent from the core Network
functionality the exess network provides
12:58 a core Network Technology independent access for mobile terminals to different
types of core networks and Network
13:07 Services either core Network domain can access any appropriate RN service
example it should be possible to access
13:15 a speech radio access Bearer from the packet switched domain the radio Access
13:23 Network consist of new network elements known as node B and Radio Network
Network controllers
13:31 rnc's node B is comparable to the base transceiver station in 2G wireless
13:38 networks RNC replaces the base station controller it provides the radio Resource
Management Handover control and
13:47 support for the connections to Circuit switched and packet switched domains the
interconnection of the network elements
13:55 in R and between RN and core network is over Lu L and LU interfaces based on
ATM
14:04 as a layer two switching technology data services run from the terminal device over
IP which in turn uses ATM as a
14:13 reliable transport with qos voice is embedded into ATM from the edge of the
network node B and is
14:22 transported over ATM out of the RNC the Lu interface is split into two
14:29 Parts circuit switched and packet switched the Lu interface is based on ATM with
voice traffic embedded on

14:37 Virtual circuits using A2 technology and IP over ATM for data traffic using a A5
14:47 technology these traffic types are switched independently to either 3G sgsn
14:54 for data or 3D MSC for voice figure shows was the 3G wireless network
15:02 architecture effect of 4G on 3G network technology nowadays in Indian
15:09 telecommunication Market 4G network is the hottest topic this excitement has been
15:17 fueled out by the launch of 4G
15:17 Reliance Geo considered as the largest investment in Indian history it is important
15:26 to note that Indian operators
15:26 invested significantly in 3G network and Spectrum from 2010
15:32 2015 slow uptake of 3G in India means a lot of that investment has not yet been
15:39 fully recovered as 4G becomes the dominant option some operators are considering
15:47 shutdown of their 3D
15:47 networks to simplify their architectures however this is happening in countries
15:55 where typically 3D networks have been in
15:55 operation for almost 15 years on the other hand developing countries like Indonesia
16:04 Thailand Etc continue to focus
16:04 on 3G even though they have launched 4G networks and are looking for a slow
16:12 migration to 4G
16:12 networks India 2 was in a similar stage of evolution until 2015 and even 5 years
16:19 after launch 3D penetration was less than 10% one key reason was that
16:26 smartphones
16:26 and 3G data pricing were both both expensive no wonder we had 300 million
16:36 internet users of which 180 million had
16:36 smartphones but only 90 million bought 3G plans network security network
16:44 security consists of the policies and
16:44 practices adopted to prevent and monitor unauthorized access misuse modification
16:53 or denial of a computer network and
16:53 network accessible resources network security involves the authorization of access
17:01 to data in the network which is
17:01 17 minutes, 1 second controlled by the network administrator users Choose Or
17:10 assigned an ID and password or other
17:10 authenticating information that allows them access to information and programs
17:16 within their
17:16 Authority network security covers a variety of computer networks both public and
17:25 private that are used in everyday
17:25 jobs conducting transactions and Communications among businesses government
17:32 agencies and
17:32 individuals networks can be private such as within a company and others which
17:32 might be open to public

17:39 access network security is involved in organizations Enterprises and other types of Institutions it does as its

17:48 title explains it secures the network as well as protecting and overseeing operations being done the most common

17:57 and simple way of prot protecting a network resource is by assigning it a unique name and a corresponding

18:04 password security involves the following three elements confidentiality to prevent unauthorized

18:12 disclosure of information to third parties this includes the disclosure of information about resources Integrity to

18:22 prevent unauthorized modification of resources and maintain the status qu it includes the Integrity of system

18:30 resources information and personal the alteration of resources like information may be caused by a desire for personal

18:39 gain or a need for Revenge availability to prevent unauthorized withholding of system resources from those who need

18:48 them when they need them communication infrastructure is a vital component of every country it could be targeted in an

18:56 attack or also with be on only destructive Purpose with the intent to steal sensitive information these days

19:04 we read nearly all the vulnerabilities that can be exported to undermine the communications back DUIs by governments

19:13 in the surveillance of mobile devices flaws in authentication protocols that allow you to very easily Pierce our

19:22 Wi-Fi networks and malware hacking used to control the VIP Communications from our PC 3G offer greater security

19:31 allowing Mutual authentication between terminals and networks with 3G network growth the mobile will play a similar

19:39 ordinary role of a computer hackers are able to penetrate mobile devices exactly in the same way they access to our

19:47 confidential data on our computer hackers can easily Spire our movements listen to our phone call read

19:56 our messages access to all our private data this third generation technology brings along with it a vast number of

20:05 vulnerabilities making it a heaven for hackers and crackers all this while very few

20:12 consumers were actually aware of the threats to 3G and each operator will need to spend 5 to 10% of its gain in

20:21 securing 3D Services suffer more security threats and it is necessary to Define new security Solutions at

20:29 different levels both at the service providers and and the handset manufacturers
the 3D networks will make

20:37 a mobile communication Network and bandwidth equivalent to a computer network
which will in turn open the

20:45 chances for cyber criminals to carry out attacks at will through the mobile
networks from a purely technological

20:53 perspective 3G networks use the Kasumi block crypto in instead of the older A5
by1 stream Cipher but also Kazumi Cipher has been identified several serious

21:00 weaknesses consider also that the increasing of connectivity means a sensible grow
of security exposure

21:08

21:15 harder to manage main security problems related to 3G networks

21:22 imsi is sent in clear text when allocating tmsi to the user

21:29 the transmission of IMI is not protected IMI is not a

21:37 security feature a user can be enticed to camp on a false

21:44 BS once the user camps on the radio channels of a false BS the user is Out

21:51 Of Reach of the paging signals of SN hijacking outgoing incoming calls in the

21:57 networks with the disabled encryption is possible the Intruder poses a man in the
middle and drops the user once the call

22:06 is set up according to etsi ts121 133 specification 3D threats could

22:15 be classified as unauthorized access to sensitive data violation of

22:22 confidentiality eavesdropping an intruder intercepts messages without detection

22:29 masquerading an intruder Hawes an authorized user into believing that they are
the legitimate system to obtain

22:38 confidential information from the user or an intruder Hawes a legitimate system
into believing that they are an

22:47 authorized user to obtain system service or confidential information traffic

22:53 analysis an intruder observes the time rate length source and destination of

23:00 messages to determine a user's location or to learn whether an important

23:07 business transaction is taking place browsing an intruder searches data storage for
sensitive information

23:15 leakage an intruder obtains sensitive information by exploiting processes with the
legitimate access to the data

23:24 inference an intruder observes a reaction from a system system by sending a query
or signal to the system for

23:32 example an intruder May actively initiate Communications sessions and then obtain
access to information

23:39 through observation of the time rate length sources or destinations of associated
messages on the radio

- 23:46 interface unauthorized manipulation of sensitive data that is violation of Integrity manipulation of messages
- 23:55 messages may be deliberately modified inserted replayed or deleted by an intruder disturbing or misusing Network
- 24:03 Services leading to denial of service or reduced availability intervention an intruder
- 24:11 May prevent a authorized user from using a service by jamming the users traffic signaling or Control Data resource
- 24:20 exhaustion an intruder May prevent an authorized user from using a service by overloading the service misuse of
- 24:28 privileges a user or a serving Network May exploit their privileges to obtain unauthorized services or
- 24:36 information abuse of services an intruder May abuse some special service or facility to gain an advantage or to
- 24:44 cause disruption to the network repudiation a user or a network denies
- 24:51 actions that have taken place unauthorized access to services Intruders can access Services by
- 25:00 masquerading as users or network entities users or network entities can
- 25:07 get unauthorized access to Services by misusing their excess rights so friends
- 25:14 this was all about 3G network and 3G network security hope to see you in the
- 25:20 next lecture till then goodbye thank you
- 25:24 [Music]
- 25:32 [Applause]
- 25:39 [Music]